

Airbnb Price Prediction Report - Paris

Generated on 2026-08-25 by the automated pipeline (*make all*).

Executive summary

The pipeline trained and compared Random Forest, XGBoost and TabPFN on Airbnb listing data for **Paris** (33881 listings). The **xgboost** model was selected for final evaluation on a held-out test set and achieved: - **RMSE: \$153** - **MAE: \$84** - **R²: 0.69**

Data

- Source: InsideAirbnb (latest available snapshot). See: <https://insideairbnb.com/get-the-data/>
- City: Paris · 33881 listings after cleaning

EDA

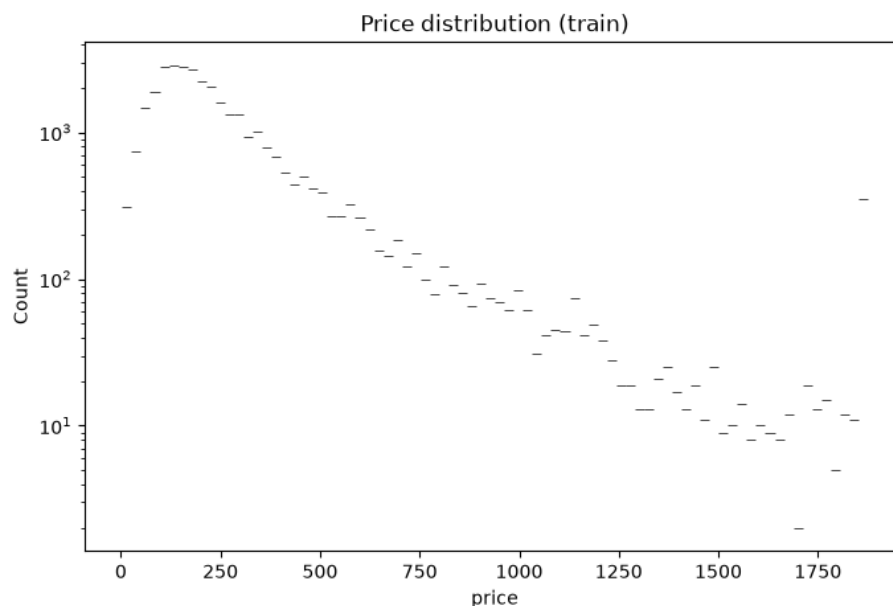
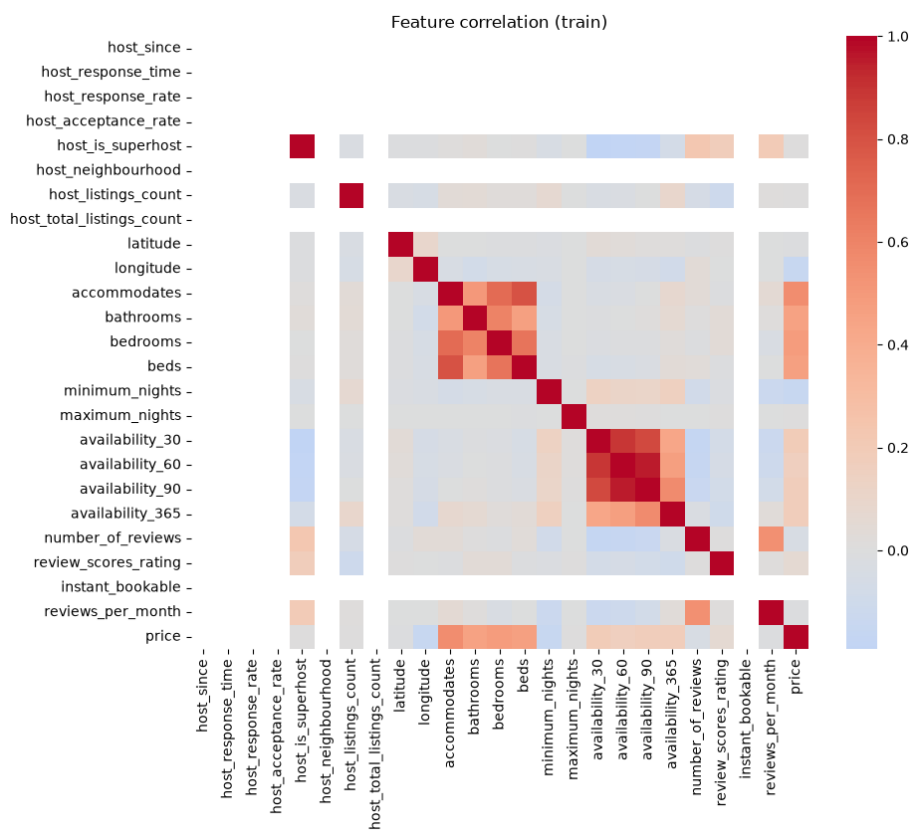
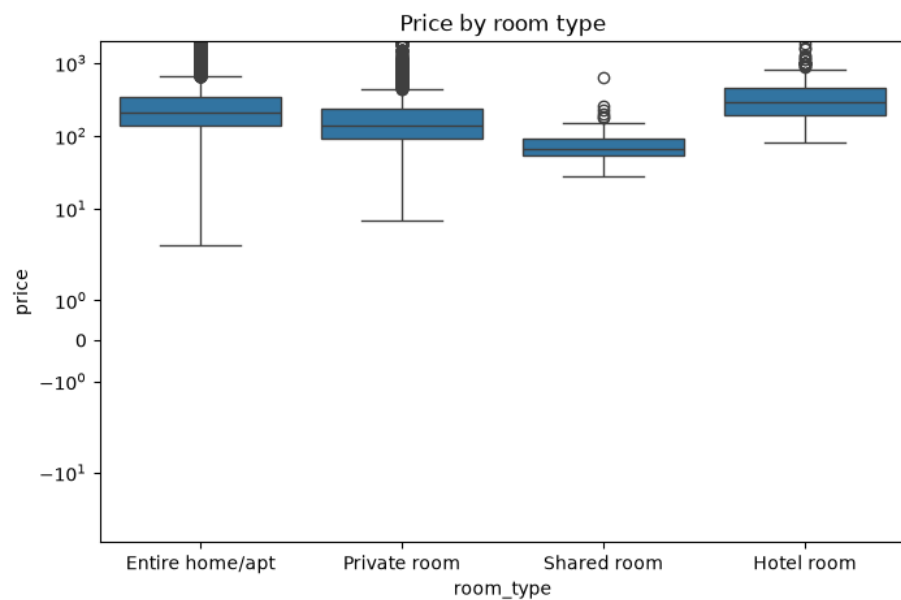


Figure 1: price_dist

- Median price: **\$206** (mean \$294)



Models & Results

Model	Validation RMSE
Random Forest	\$173
XGBoost	\$158
TabPFN (capped sample, not directly comparable)	\$175

Holdout results (best model): see `artifacts/Paris/holdout_results.csv`.

Top features

Feature	Importance
bathrooms	0.193
accommodates	0.084
bedrooms	0.078
availability_90	0.061
minimum_nights	0.049
property_type_Room in boutique hotel	0.044
room_type_Entire home/apt	0.043
property_type_Private room in rental unit	0.028
property_type_Room in hotel	0.026
room_type_Private room	0.026

Recommendations

- The best-performing model was **xgboost** (holdout RMSE \$153, MAE \$84, R^2 0.69).
- On validation, **XGBoost** beat Random Forest by 9% on RMSE (\$158 vs \$173). A simple ensemble or blending the two may squeeze out further gains.
- Pricing is most strongly driven by **bathrooms, accommodates, bedrooms** — i.e. bathrooms (a key listing attribute); accommodates (how many guests a listing sleeps); bedrooms (the number of bedrooms). Hosts can use these levers when setting nightly rates.
- Accuracy caveat: the holdout RMSE of \$153 is **75% of the median listing price (\$206)**, so predictions for typical listings are expected to be off by roughly $\pm\$153$.
- Next steps: add richer features (neighbourhood_cleansed, amenities, review scores breakdowns), train on a log-transformed target to tame high-end outliers, run a deeper hyperparameter search, and extend the pipeline to more cities.

Automated pipeline: fetch → preprocess → EDA → train → evaluate → report.
*Code in **src/**, config in **configs/base.yaml**.*